

News Credibility Analysis Using Logistic Regression

Team Name: X

Team Members:

Advik Khandelwal, Sahil Kumar, Harsh Ahalawat, Meet Kumar

February 23, 2026

Abstract

This project presents a machine learning based News Credibility Analysis system that predicts whether a news article is real or fake. Natural Language Processing (NLP) techniques and Logistic Regression are used to classify textual news content. The system performs preprocessing, feature extraction using TF-IDF, and classification with high accuracy.

1 Introduction

The rapid spread of misinformation through online platforms has made automatic fake news detection an important research problem. Manual verification of news articles is time-consuming, motivating the use of machine learning techniques for automated credibility analysis.

This project builds a model that classifies news articles as real or fake based on textual content.

2 Dataset

The dataset was obtained using KaggleHub and consists of news articles containing the following fields:

- Title
- Text content
- Label (Real or Fake)

The title and article text were combined to form the final input content. Missing values were removed before training.

3 Data Preprocessing

Text preprocessing was performed using Natural Language Processing techniques:

- Conversion of text to lowercase
- Removal of URLs and special characters
- Retaining alphanumeric characters
- Stopword removal using NLTK

A cleaned text column was generated and used for feature extraction.

4 Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) vectorization was used to convert text into numerical features.

The configuration used:

- Maximum features: 20,000
- N-gram range: (1,2) (unigrams and bigrams)
- Minimum document frequency: 5
- Maximum document frequency: 0.7
- Sublinear term frequency scaling enabled

5 Model Training

The dataset was split into training and testing sets using an 80:20 ratio with stratified sampling.

A Logistic Regression classifier was trained with:

- Maximum iterations: 2000
- Class weight: balanced
- Regularization parameter (C): 2.0

Logistic Regression was chosen due to its efficiency and strong performance in binary text classification problems.

6 Model Evaluation

Model performance was evaluated using precision, recall, F1-score, and confusion matrix metrics.

6.1 Classification Report

	precision	recall	f1-score	support
0	0.97	0.96	0.96	7006
1	0.96	0.97	0.97	7302
accuracy			0.96	14308
macro avg	0.96	0.96	0.96	14308
weighted avg	0.96	0.96	0.96	14308

6.2 Confusion Matrix

```
[[6726  280]
 [ 232 7070]]
```

The model achieves an overall accuracy of approximately 96%, indicating strong predictive capability and balanced performance across both classes.

7 Model Deployment

The trained Logistic Regression model and TF-IDF vectorizer were saved using Python's pickle module. These saved models were later integrated into a Streamlit web application that allows users to input news text and receive real-time credibility predictions.

8 Conclusion

This project demonstrates the effectiveness of machine learning and NLP techniques for fake news detection. Logistic Regression combined with TF-IDF feature extraction provides accurate and computationally efficient classification results.

9 Future Work

Future improvements may include:

- Using deep learning models such as LSTM or Transformers
- Expanding dataset diversity

- Deploying the system on cloud platforms
- Adding explainable AI visualizations